

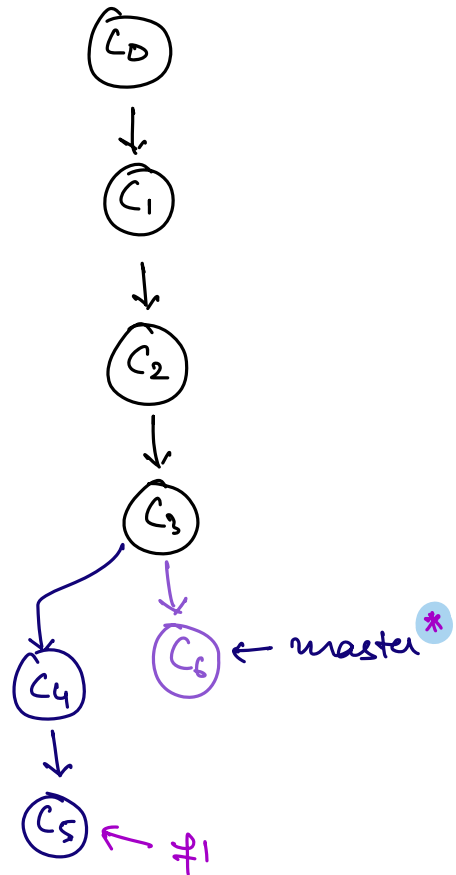
Agenda.

- Git branching
- Git Merge
- Merge Conflicts
- Git Rebase.

Git Branch.

⇒ Parallel line of development.

⇒ Branch is a pointer pointing to the latest commit in that line of development.



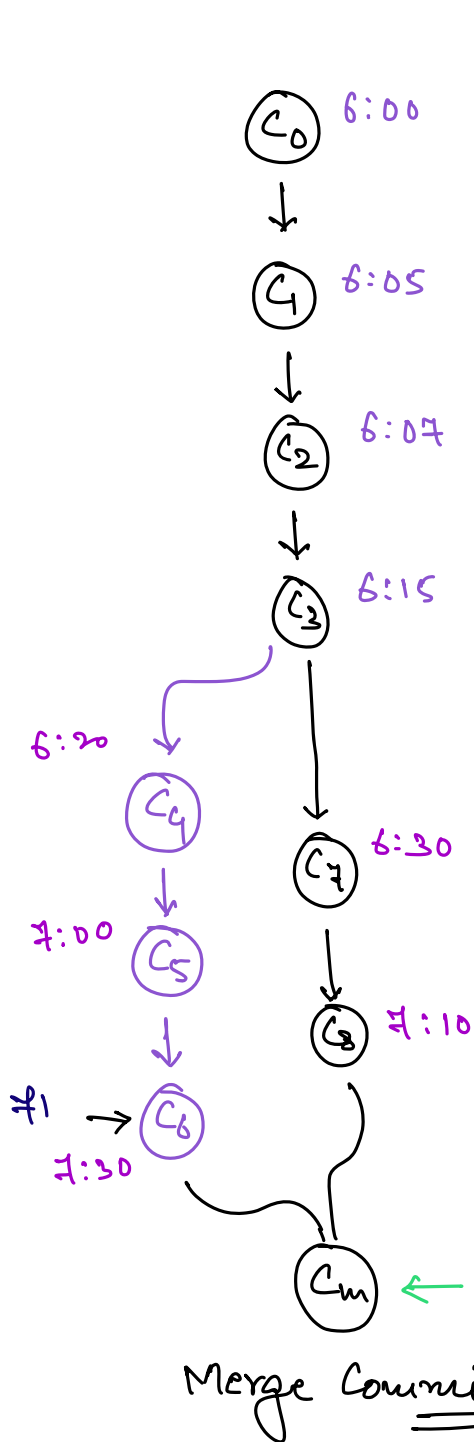
git branch -b f1

git checkout f1



git checkout -b f1.

#



Bring changes of one branch to another branch.

Merge.

⇒ Merge #1 into master.

1) git checkout master

2) git merge #1.

⇒ git log ⇒

- C6
- C8
- C5
- C7
- C4
- C3
- C2
- C1
- C0

⇒ Not a clean history.

A.txt

master

```
1
2
3
4
5
6
7
```

#1

A.txt

```
1 0
2 0
3 0
4
5
6
7
```

A.txt

master

```
1 1
2 2
3 3
4
5
6
7
```

→ Merge #1 into main.

→ Merge Conflict.

A.txt

```
1 0
2 0
3 0
4
5
6
7
```

A.txt

```
1 1
2 2
3 3
4
5
6
7
```

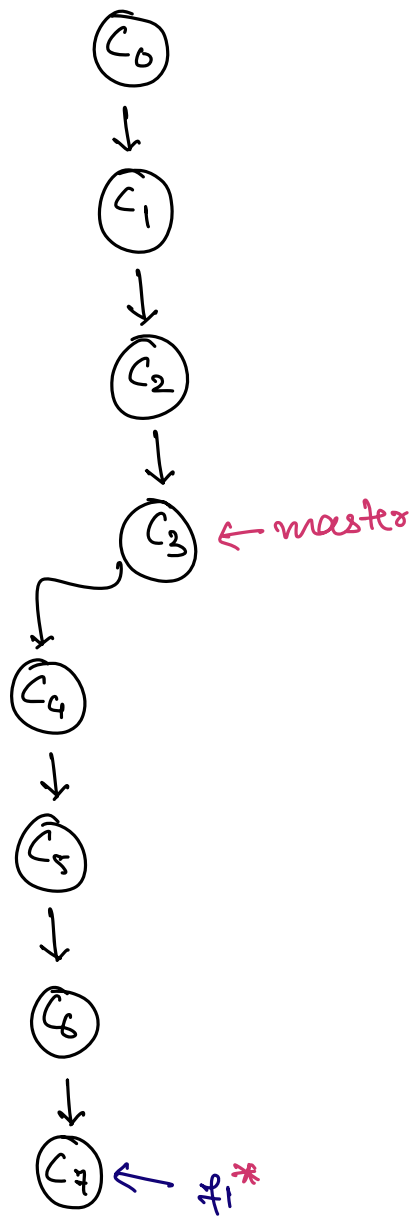
A.txt

```
1 1 1 0
2 2 2 0
3 3 3 0
4
5
6
7
```

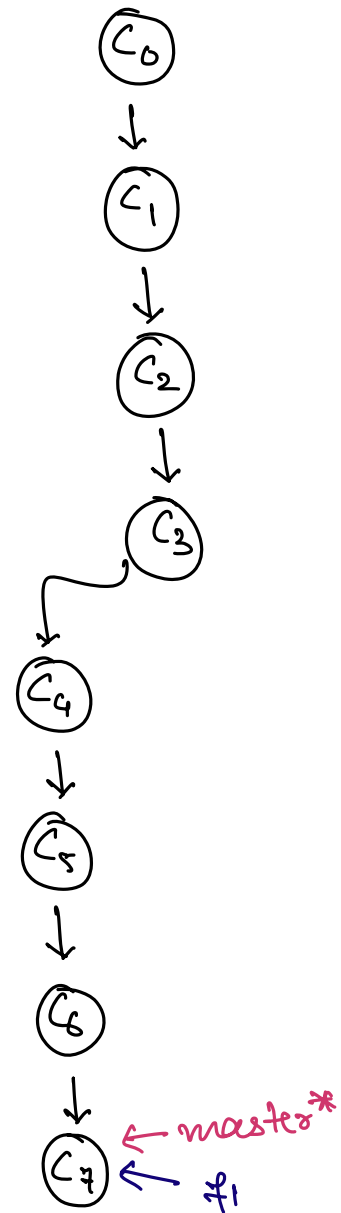
...

#

Merge #1 into main

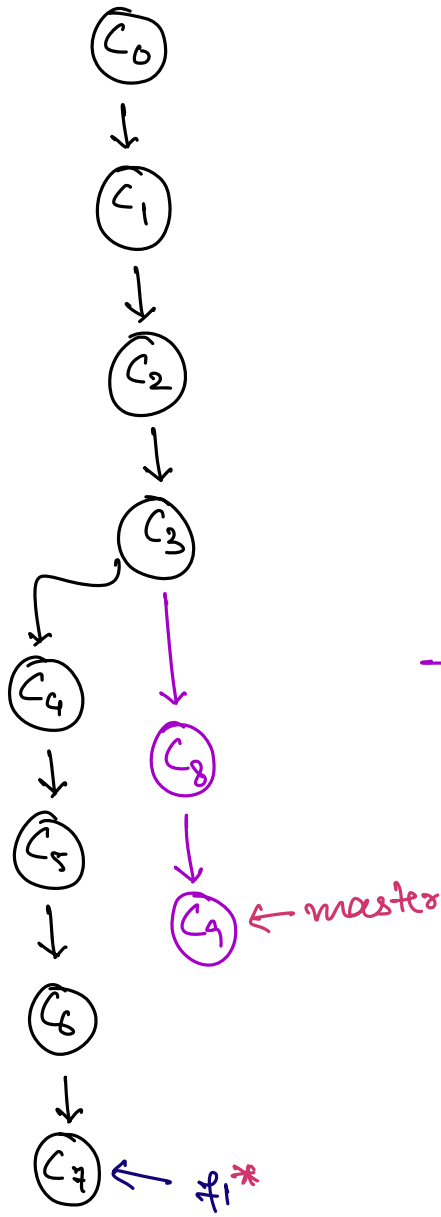


Merge
#1 into
master.
=

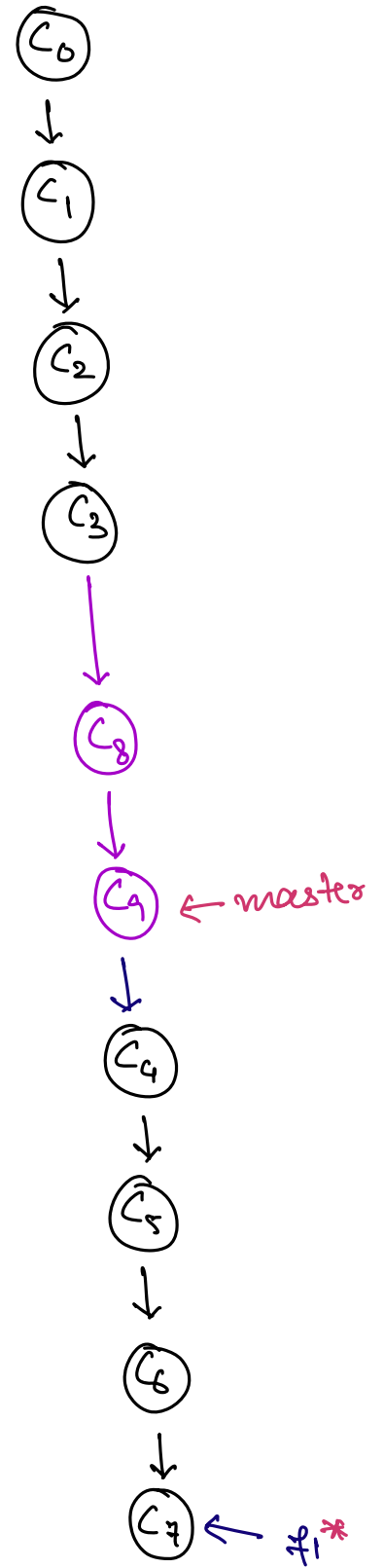


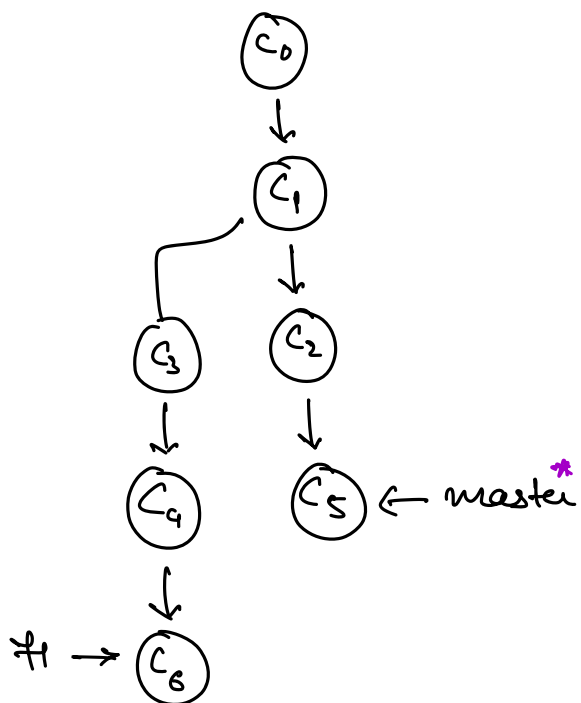
⇒ git log
↪ clean git history.

Rebase.

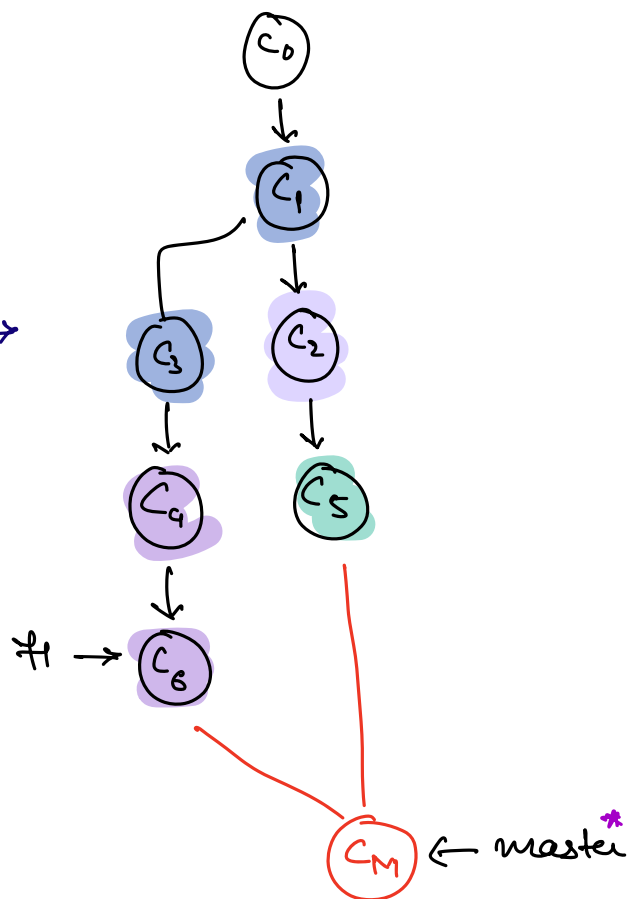


Rebase f1
onto main

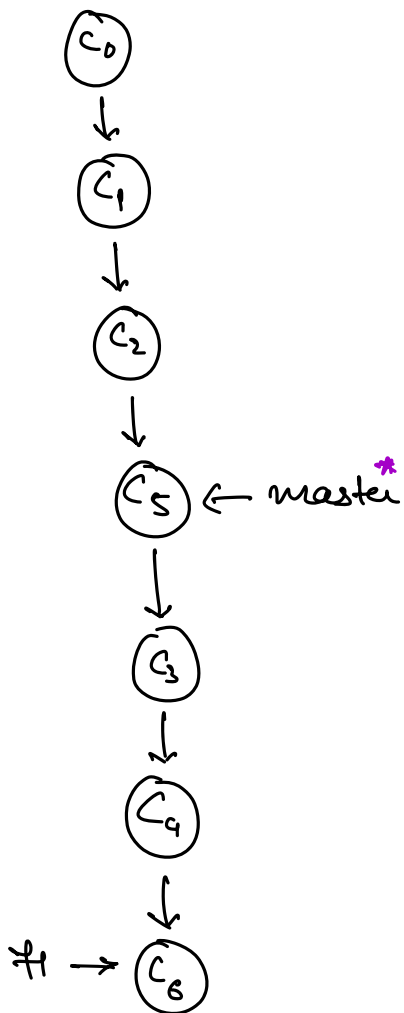




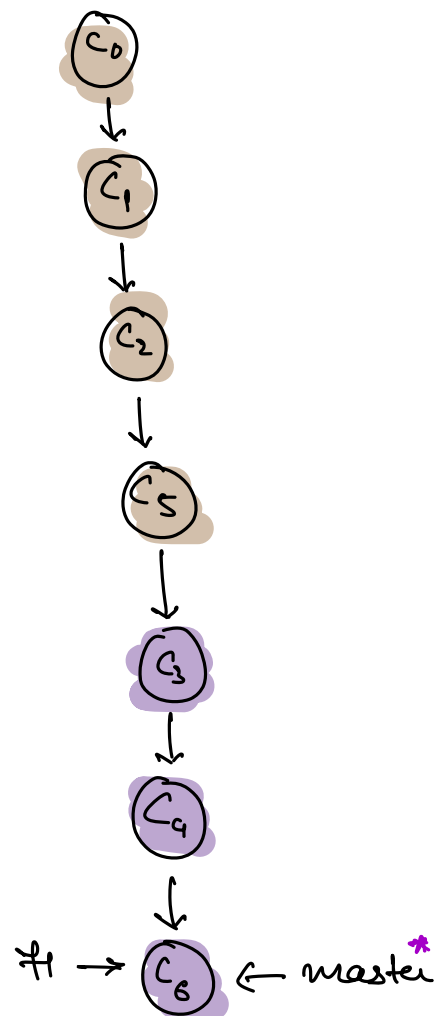
Merge f1
into master



Rebase f1
onto master



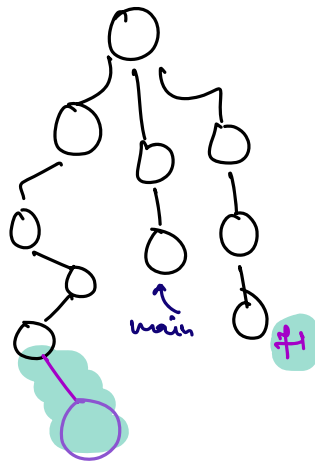
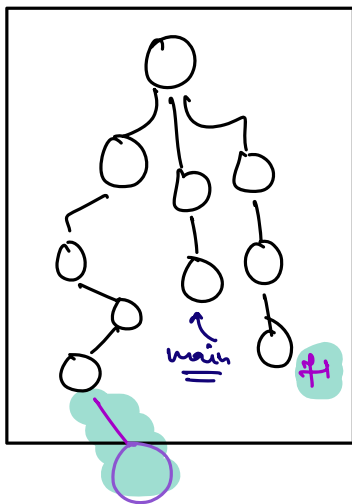
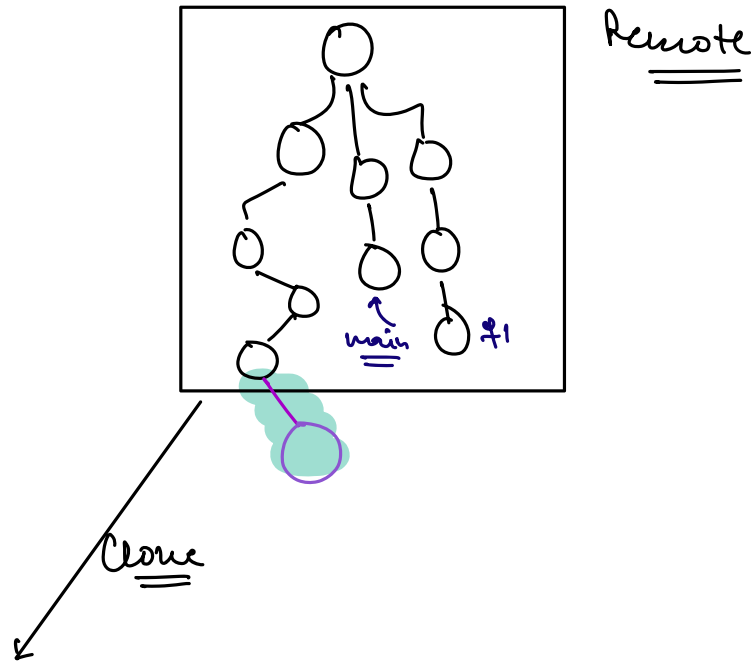
Merge f1
into master



Merge

(vs)

Rebase + Merge



Read-only

→ git fetch: fetches the changes from remote to local read-only branch.

Fetch
+
Merge } git pull.

Fetch
+
Rebase
+
Merge } git pull --rebase.